Sample title

Unpaired Point Cloud Translation with Diffusion Models for Space Charge Correction in Time Projection Chambers

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(Dated: 7 April 2026)

[Abstract goes here]

Keywords: Suggested keywords

I. INTRODUCTION

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II. RELATED WORK

A. Diffusion Models for Point Clouds

Diffusion probabilistic models (DPMs) generate data by learning to reverse a noise-corruption process, and have achieved strong results across generative tasks in both images[1,2] and scientific data. In particle physics, DPMs have been applied to point cloud generation for jet simulation, where models such as PC-JeDi[3], PC-Droid[4], and EPiC-FM[5] have established state-of-the-art performance for generating jet particle clouds with permutation equivariance. Diffusion models have also been used for calorimeter shower simulation, both on voxelized representations and as geometry-independent point clouds in the CaloClouds series[6]. More recently, point cloud diffusion has been applied to nuclear and heavy-ion physics: Araz et al. used Point Edge Transformers for event generation at the Electron-Ion Collider[7], and Omana Kuttan et al. built on the Luo and Hu architecture—the same backbone we use—to generate heavy-ion collision events[8]. For TPC data, Imani et al. developed a score-based diffusion model for LArTPC image generation[9], and Go et al. applied DDPMs to sPHENIX whole-event simulation[10]. Our work differs from these in that we use diffusion models not for generation but for unpaired translation between domains.

B. Unpaired Translation

Unpaired image-to-image translation was introduced by Zhu et al. with CycleGAN[11], which uses cycle consistency loss to learn domain mappings without paired data. Diffusion-based alternatives have since emerged: UNIT-DDPM[12] trains a joint diffusion model across domains, EGSDE[13] uses energy functions to guide SDE-based translation, and DDIB[14] performs translation by encoding and decoding through a shared Gaussian latent space via two independently trained models. Most relevant to our work is CycleDiffusion by Wu and De la Torre[15], which defines a latent space for stochastic DPMs and introduces the DPM-Encoder for mapping inputs to an exactly reconstructable latent code—the method we adapt for point clouds.

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In physics, Huang et al.[16] applied several unpaired image translation methods to bridge the simulation–experiment gap in LArTPC detector responses, finding UVCGAN[17] to be most effective. In the point cloud domain, unpaired translation remains largely unexplored: Zheng et al.[18] and Li and Lin[19] proposed non-diffusion architectures for translating 3D object point clouds, and Zhang et al.[20] recently introduced the first diffusion-based method for unpaired point cloud completion. To our knowledge, our work is the first to apply diffusion-based unpaired translation to point cloud detector data.

C. Space Charge Effects in Time Projection Chambers

In active-target TPCs, the beam passes directly through the detector gas, producing a column of positive ions whose slow drift creates electric field distortions that deflect electron trajectories and distort reconstructed tracks. [Space charge effect discussion.]

On the machine learning side, the ALICE experiment at CERN uses convolutional neural networks for real-time space charge distortion correction during Run 3[21], though this relies on supervised training with paired data. In the AT-TPC context, machine learning has been applied to event classification—Kuchera et al.[22] used CNNs for track classification, and Dey et al.[23] recently demonstrated PointNet-based classification of fission events in AT-TPC data—but not yet to detector effect correction. Our approach offers a data-driven alternative that requires no paired data and no detailed physical modeling of the space charge distribution.

III. METHODS

A. Problem Formulation

We cast the correction of detector effects as an unpaired point cloud translation task. Given two datasets $\mathcal{D}_X$ and $\mathcal{D}_Y$ comprising point clouds, where $\mathcal{D}_X$ represents events affected by a detector effect and $\mathcal{D}_Y$ represents events without, we seek to learn mappings $G : \mathcal{D}_X \to \mathcal{D}_Y$ and $H : \mathcal{D}_Y \to \mathcal{D}_X$ that can transform events from one domain into events that resemble members of the other domain. Each point cloud consists of N points with d features. In our case, $N = 512$ points with $d = 4$ features (three spatial coordinates and charge). Crucially, no one-to-one correspondence between the elements of $\mathcal{D}_X$ and $\mathcal{D}_Y$ is required to learn these mappings.

B. Point Cloud Diffusion Models

Our framework builds on the point cloud diffusion model introduced by Luo and Hu[24]. A diffusion probabilistic model (DPM) consists of two processes: a forward process that gradually corrupts data with noise, and a reverse process that learns to denoise. In the point cloud setting, each point in the cloud is treated as an independent particle drawn from a point distribution conditioned on a shared shape latent $\mathbf{z}$ that encodes the global geometry. The forward and reverse processes operate independently on each point, with all points sharing the same $\mathbf{z}$.

a. Forward process. Given an input point cloud $\mathbf{X}^{(0)} = \{\mathbf{x}_i^{(0)}\}_{i=1}^N$, the forward process defines a Markov chain that progressively adds Gaussian noise to each point over T steps:

$$q(\mathbf{x}_i^{(t)} \mid \mathbf{x}_i^{(t-1)}) = \mathcal{N}\Big(\mathbf{x}_i^{(t)} \;\Big|\; \sqrt{1-\beta_t}\,\mathbf{x}_i^{(t-1)},\ \beta_t \mathbf{I}\Big), \quad t = 1, \ldots, T \tag{1}$$

where $\beta_1, \ldots, \beta_T$ is a variance schedule controlling the noise added at each step. A useful property of this formulation is that one can sample any intermediate noisy point directly

without iterating through all prior steps. *[Add the closed-form sampling formula.]* After sufficiently many steps, $\mathbf{x}_i^{(T)}$ is approximately distributed as standard Gaussian noise.

b. Reverse process. The reverse process learns to recover $\mathbf{x}_i^{(0)}$ from $\mathbf{x}_i^{(T)}$ by iteratively denoising each point, conditioned on the shape latent $\mathbf{z}$. Each reverse step is modeled as a Gaussian transition:

$$p_\theta(\mathbf{x}_i^{(t-1)} \mid \mathbf{x}_i^{(t)}, \mathbf{z}) = \mathcal{N}\Big(\mathbf{x}_i^{(t-1)} \;\Big|\; \boldsymbol{\mu}_\theta(\mathbf{x}_i^{(t)}, t, \mathbf{z}),\ \beta_t \mathbf{I}\Big) \tag{2}$$

where $\boldsymbol{\mu}_\theta$ is a learned mean estimator implemented as a neural network parameterized by θ. Since each point is diffused independently, the joint probability of the entire point cloud factorizes as a product over points, with all points sharing the same shape latent:

$$p_\theta(\mathbf{X}^{(0:T)} \mid \mathbf{z}) = \prod_{i=1}^{N} p_\theta(\mathbf{x}_i^{(0:T)} \mid \mathbf{z}). \tag{3}$$

c. Training objective. The model is trained by maximizing a variational lower bound on the log-likelihood of the point cloud[24]. *[I need to look at the papers more.]*

d. Architecture. The encoder is a PointNet[25] that maps the input point cloud $\mathbf{X}^{(0)}$ to parameters of a shape latent distribution $q_\phi(\mathbf{z} \mid \mathbf{X}^{(0)})$. In the auto-encoding setting used in our framework, the shape latent $\mathbf{z}$ is sampled from this distribution and passed to the decoder, which implements the reverse diffusion process. The PointNet architecture achieves permutation invariance through per-point transformations followed by a symmetric max-pooling operation, ensuring that the shape latent is independent of point ordering.

C. CycleDiffusion for Point Clouds

Our system combines CycleDiffusion[15], an unpaired image translation framework, with the point cloud diffusion model of Luo and Hu[24]. CycleDiffusion relies on a striking property of diffusion models: they produce "uniquely identifiable encodings"[2], meaning that diffusion models trained on the same data produce close to identical outputs when primed with the same randomly sampled input, regardless of model architecture or training procedure[2]. Remarkably, Khrulkov et al.[26] found that this alignment of latent representations persists even when models are trained on entirely different datasets—high-level semantic features such as pose, shape, and spatial structure are preserved when the same latent code is decoded by networks modeling completely different distributions. This suggests a natural approach to unpaired translation between domains: we independently train two diffusion models on the two domains of interest $\mathcal{D}_X$ and $\mathcal{D}_Y$, encode a source point cloud using its in-domain encoder, and then decode the resultant latent representation with the other diffusion model. This, in effect, is CycleDiffusion[15]. We leverage Luo and Hu's point cloud diffusion architecture to adapt this idea to the translation of point clouds[24]. We discuss the relevant technical details in the rest of this section.

a. Latent space of stochastic diffusion models. We now make the above intuition precise. Wu and De la Torre[15] showed that stochastic DPMs admit a well-defined latent space despite the noise injected at each denoising step. The key idea is to treat the collection of all noise vectors used during generation as the latent code. Specifically, given the reverse process

$$\mathbf{x}_{t-1} = \boldsymbol{\mu}_\theta(\mathbf{x}_t, t) + \sigma_t \boldsymbol{\epsilon}_t, \quad \boldsymbol{\epsilon}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \quad t = T, \ldots, 1 \tag{4}$$

the latent code is defined as $\mathbf{z} := \mathbf{x}_T \oplus \boldsymbol{\epsilon}_T \oplus \cdots \oplus \boldsymbol{\epsilon}_1$, where $\oplus$ denotes concatenation. Once this latent code is fixed, the output of the stochastic DPM is fully determined—the model becomes a deterministic map from $\mathbf{z}$ to an output. This is what enables unpaired translation: if two independently trained DPMs share the same latent code, they will produce structurally similar outputs that differ only in domain-specific characteristics.

b. DPM-Encoder. To encode an existing point cloud into this latent space, Wu and De la Torre[15] proposed the DPM-Encoder. Their key insight was to retain the noise added to the data during each step of the forward diffusion process, together with the final noisy output $\mathbf{X}^{(T)}$, and to treat the concatenation of these quantities as the latent encoding of the original input. Given an input $\mathbf{X}^{(0)}$ and a trained DPM, the DPM-Encoder leverages the fact that the forward process $q(\mathbf{x}_i^{(1:T)} \mid \mathbf{x}_i^{(0)})$ is predefined. One samples a forward trajectory $\mathbf{x}_i^{(1)}, \ldots, \mathbf{x}_i^{(T)}$ from the forward process and then computes the effective noise at each reverse step by definition:

$$\boldsymbol{\epsilon}_t = \frac{\mathbf{x}_i^{(t-1)} - \boldsymbol{\mu}_\theta(\mathbf{x}_i^{(t)}, t, \mathbf{z})}{\sigma_t}, \quad t = T, \ldots, 1. \tag{5}$$

The latent encoding is then $\boldsymbol{\epsilon}_X = \boldsymbol{\epsilon}_T \oplus \boldsymbol{\epsilon}_{T-1} \oplus \cdots \oplus \boldsymbol{\epsilon}_1$ together with $\mathbf{X}^{(T)}$. An important property of this encoding is perfect reconstruction: feeding $\mathbf{X}^{(T)}$ through the reverse diffusion process using the stored noise vectors $\boldsymbol{\epsilon}_t$ at each step, rather than sampling fresh noise, recovers $\mathbf{X}^{(0)}$ exactly. It is worth noting, however, that the encoding itself is stochastic—different runs of the forward process produce different trajectories and therefore different latent codes—though each individual code permits exact reconstruction.

c. Adaptation for point clouds. While the DPM-Encoder leads to impressive results for unpaired image translation[15], it cannot be applied directly to point clouds. Unlike image data, where the spatial structure of the input is implicitly preserved by the fixed pixel grid, point cloud events are permutation invariant—any reordering of the points represents the same underlying geometry. In the Luo and Hu[24] framework, noise is added independently to each point, so the noise vectors $\boldsymbol{\epsilon}_X$ alone do not encode the global topological structure of the point cloud, such as the number, orientation, and spatial extent of particle tracks. We address this by augmenting the DPM-Encoder with an additional shape latent computed by a PointNet encoder[25] (see Fig. 1). This enables the encoding of topological information about the original point cloud $\mathbf{X}^{(0)}$ in the latent representation, prior to decoding and translation via the reverse diffusion process. The modified DPM-Encoder thus produces a latent encoding consisting of three components: the final noisy point cloud $\mathbf{X}^{(T)}$, the noise vectors $\boldsymbol{\epsilon}_X$, and the shape latent $\mathbf{z}$. During decoding, the reverse diffusion process is conditioned on $\mathbf{z}$, which provides a permutation-invariant structural blueprint of the original point cloud, while the noise vectors supply the step-by-step denoising schedule.

d. Translation procedure. An overview of the inference pipeline is shown in Fig. 1. We train two independent point cloud DPMs, one for each domain $\mathcal{D}_X$ and $\mathcal{D}_Y$, following the architecture described in Section III B. We denote the trained encoder and decoder for domain $\mathcal{D}_X$ as Enc_X and Dec_X, and similarly Enc_Y and Dec_Y for domain $\mathcal{D}_Y$. Training is performed independently—the two models share no parameters and are not jointly optimized.

To translate an event $\mathbf{X}^{(0)}$ from domain $\mathcal{D}_X$ to domain $\mathcal{D}_Y$, we proceed as follows:

1. **Encode shape latents.** Process $\mathbf{X}^{(0)}$ through both PointNet encoders to obtain latent representations in each domain:

$$\mathbf{z}_X = \mathrm{Enc}_X(\mathbf{X}^{(0)}), \quad \mathbf{z}_Y = \mathrm{Enc}_Y(\mathbf{X}^{(0)}). \tag{6}$$

2. **DPM-Encode.** Pass $\mathbf{z}_X$ and $\mathbf{X}^{(0)}$ to the modified DPM-Encoder, which simulates the forward diffusion process for T steps using model X, producing the fully noised point cloud $\mathbf{X}^{(T)}$ and the sequence of noise vectors $\boldsymbol{\epsilon}_X = \boldsymbol{\epsilon}_T \oplus \cdots \oplus \boldsymbol{\epsilon}_1$.

3. **Decode in target domain.** Initialize $\mathbf{Y}'^{(T)} = \mathbf{X}^{(T)}$ and run the reverse diffusion process using the decoder of model Y, conditioned on $\mathbf{z}_Y$ and guided by $\boldsymbol{\epsilon}_X$. Each denoising step is computed as:

$$\mathbf{Y}'^{(t-1)} = \boldsymbol{\mu}_Y(\mathbf{Y}'^{(t)}, t, \mathbf{z}_Y) + \sigma_t \boldsymbol{\epsilon}_t, \quad t = T, \ldots, 1 \tag{7}$$

where $\boldsymbol{\mu}_Y$ is the learned mean estimator of Dec_Y.

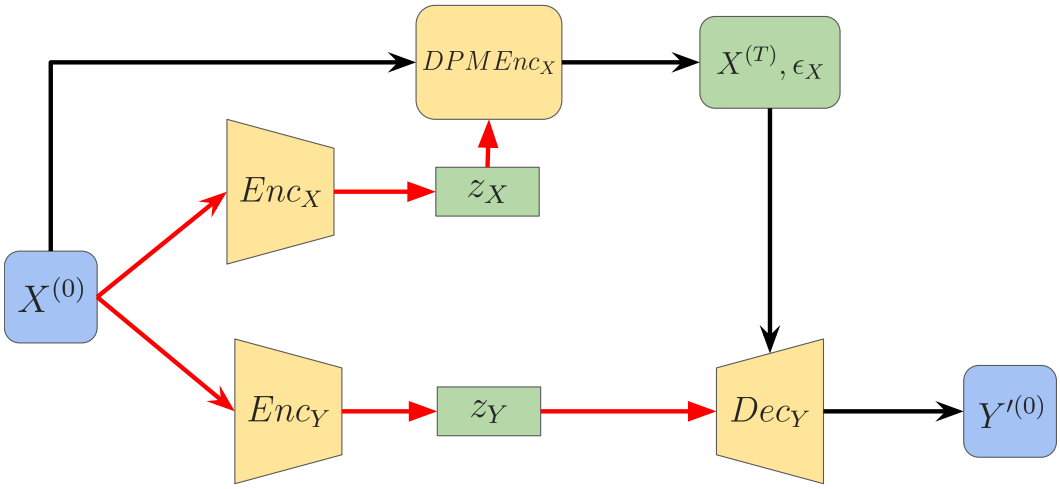

FIG. 1. Schematic depicting the unpaired translation model at inference time. To translate an event $\mathbf{X}^{(0)}$ from domain $\mathcal{D}_X$ to domain $\mathcal{D}_Y$, the event is processed through both domain encoders to obtain shape latents $\mathbf{z}_X$ and $\mathbf{z}_Y$. The modified DPM-Encoder uses $\mathbf{z}_X$ and $\mathbf{X}^{(0)}$ to produce the noised point cloud $\mathbf{X}^{(T)}$ and noise vectors $\boldsymbol{\epsilon}_X$. The target domain decoder Dec_Y then performs reverse diffusion conditioned on $\mathbf{z}_Y$ and $\boldsymbol{\epsilon}_X$ to produce the translated output $\mathbf{Y}'^{(0)}$.

The final output $\mathbf{Y}'^{(0)}$ is the translation of $\mathbf{X}^{(0)}$ into domain $\mathcal{D}_Y$. The structural identity of the original event is preserved through the shared noise vectors $\boldsymbol{\epsilon}_X$, while the target domain characteristics are imposed by the decoder Dec_Y and the shape latent $\mathbf{z}_Y$. To perform translation in the reverse direction ($\mathcal{D}_Y \to \mathcal{D}_X$), one simply swaps the roles of X and Y in the above procedure.

D. Training

The two DPMs are trained independently on their respective domains. No joint training, cycle consistency loss, or adversarial objective is used—each model simply learns to generate point clouds from its own domain via the standard diffusion training objective described in Section III B.

Both models use the encoder-decoder architecture described above, modified to accept four-dimensional point cloud data (three spatial coordinates and charge). The PointNet encoders map each point cloud to a latent vector $\mathbf{z} \in \mathbb{R}^{256}$, and the reverse diffusion process operates over $T = 256$ steps. Key hyperparameters are summarized in Table I.

TABLE I. Hyperparameters used during training of the DPMs.

Hyperparameter	Value
Initial learning rate	
Final learning rate	
Optimization algorithm	Adam
Batch size	
Number of training iterations	
Latent dimension (k)	
Diffusion steps (T)	

Model checkpoints are saved periodically during training, and the checkpoint with the lowest validation loss is selected for translation.

IV. APPLICATION: SPACE CHARGE CORRECTION IN THE AT-TPC

A. The Active-Target Time Projection Chamber

B. Space Charge Effects

C. Simulation Datasets

D. Data Preprocessing

V. EXPERIMENTS AND RESULTS

A. Evaluation Metrics

B. Translation Quality

C. Vertex Analysis

D. Reconstruction Quality

VI. DISCUSSION

VII. CONCLUSION

ACKNOWLEDGMENTS

Appendix A: Appendixes

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